

Probabilistic Collision Cone Control Barrier Function

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Abstract

Safety constitutes a fundamental requirement for autonomous systems operating in dynamic and uncertain environments. Control Barrier Functions (CBFs) have emerged as a powerful tool for enforcing safety constraints in real time; however, most existing approaches rely on the assumption of perfect state information, which is rarely satisfied in practice due to sensor noise and perception errors. In this work, a Probabilistic Collision Cone Control Barrier Function (PCC-CBF) is introduced as a novel framework explicitly accounting for uncertainty in the relative position and velocity of obstacles by modeling them as Gaussian random variables. The classical collision cone condition is reformulated as a chance constraint, ensuring that the probability of collision remains below a specified risk threshold. Tractable conditions for chance-constrained safety are derived, and the effectiveness of the PCC-CBF framework is demonstrated through simulations on unicycle and bicycle models, as well as preliminary real-world experiments. The obtained results confirm that the proposed approach enables safe and efficient navigation under uncertainty, providing a practical solution for robust autonomous operation.

1 Introduction

Ensuring safety is a fundamental requirement in the design of autonomous systems, particularly in the context of mobile robots and self-driving vehicles operating in uncertain and dynamic environments. When interacting with other traffic participants or moving obstacles, collision avoidance must be guaranteed without compromising the ability to reach a desired goal.

Among various approaches to safety-critical control, *Control Barrier Functions* (CBFs) have emerged as a powerful tool for enforcing forward invariance of safe sets in control-affine systems [2]. CBFs provide a formal mechanism for encoding safety constraints that can be enforced through online quadratic programming while maintaining system performance. However, the majority of existing CBF-based methods assume deterministic knowledge of the state and environment, which is rarely available in practice due to sensor noise and imperfect perception.

For collision avoidance, the notion of the

Collision Cone (CC) has been widely used to characterize dangerous relative motions between agents [5]. The CC provides a geometric condition to determine whether two objects are on a collision course, and several recent works have combined this concept with CBFs to ensure collision-free navigation [6]. Yet, these approaches still rely on deterministic estimates of relative position and velocity, making them sensitive to uncertainty. As perception in real-world scenarios is inherently noisy, such methods may yield overly conservative or unsafe behavior.

To address this limitation, in this work a *Probabilistic Collision Cone Control Barrier Function* (PCC-CBF) is proposed. The key idea is to extend the deterministic CC-CBF formulation by explicitly modeling uncertainty in relative position and velocity as Gaussian random variables (in most practical applications moving objects are estimated by Extended Kalman Filter which provides observed object state estimation with covariance matrix). This allows the safety condition to be reformulated as a chance

constraint, ensuring that the probability of collision remains below a specified risk threshold. The resulting probabilistic safety certificates can be integrated into a quadratic programming framework for real-time safe control.

The main contributions of this paper are threefold:

- introduction of a probabilistic formulation of Collision Cone Control Barrier Functions that accounts for uncertainty in perception,
- derivation of tractable conditions for chance-constrained safety by exploiting Gaussian approximations of state uncertainty,
- demonstration of the effectiveness of the proposed PCC-CBF approach in simulation studies on unicycle and bicycle models.

The remainder of this paper is organized as follows: Section 2 reviews related work about existing approaches to collision avoidance. Section 3 introduces the classical CBF framework and its integration with collision cones. Section 4 presents the proposed probabilistic extension. Simulation results are shown in Section 5, followed by conclusions in Section 6.

2 Literature Review

The field of safety-critical control for autonomous systems is rapidly evolving, with a strong focus on formal methods for real-time collision avoidance. This section reviews the core concepts relevant to this work, specifically Control Barrier Functions (CBFs), Collision Cone (CC) methods, and probabilistic safety guarantees in the presence of uncertainty.

Control Barrier Functions (CBFs): Control Barrier Functions (CBFs) have emerged as a foundational tool for formally guaranteeing the forward invariance of a safe set in control-affine nonlinear systems. The seminal work by [2] established the framework, showing how safety constraints can be encoded as a linear

inequality in the control input, enabling their enforcement through real-time Quadratic Programming (QP). This method ensures that if an initial state is safe, all subsequent states will remain safe. Subsequent extensions have broadened the applicability of CBFs to complex scenarios, including adaptive systems, systems with input constraints, and multi-agent coordination problems [1, 3].

A significant limitation of classical CBF approaches, however, is their reliance on a deterministic model and the assumption of perfect state information. This disconnect from the reality of noisy sensors and imperfect state estimation in real-world scenarios motivates the need for robust and probabilistic extensions.

Collision Cone Methods: For dynamic collision avoidance, the geometric concept of the Collision Cone (CC), originally developed in robotics and aerospace [?], is widely used. The CC defines a region in the relative velocity space where, given the current relative position, a collision is inevitable if the relative velocity is maintained.

Integrating this concept with CBFs has led to the development of Collision Cone Control Barrier Functions (CC-CBFs). These methods, exemplified by works such as [6] and [4], utilize the relative geometry and dynamics to construct a less conservative safety constraint compared to simple distance-based CBFs. By explicitly linking position and velocity in the barrier function definition, CC-CBFs allow for more aggressive and dynamically appropriate maneuvers.

Despite their performance advantages, existing CC-CBF formulations, like their deterministic CBF counterparts, generally assume a perfectly known relative state, making them brittle when faced with the inherent noise and uncertainty of perception systems.

Probabilistic Safety and Uncertainty: To bridge the gap between deterministic theory and real-world implementation, research has increasingly focused on probabilistic safety guarantees. These methods explicitly account for uncertainty—often modeled as Gaussian noise arising from sensors (e.g., Extended Kalman Filter outputs)—by reformulating the safety condition

as a chance constraint.

Key advancements in this area include Chance-Constrained CBFs (CC-CBFs) [?] and Stochastic Safety Certificates [?]. These approaches typically employ Gaussian approximations and linearization techniques to derive a tractable, deterministic inequality from the probabilistic constraint. This allows the system to guarantee safety with a user-defined probability $1 - \nu$, where ν is the acceptable risk level.

While these works successfully incorporate uncertainty into the general CBF framework, a gap remains: there is a lack of methods that explicitly combine the geometric, dynamic efficiency of the Collision Cone with the formal robustness of Chance-Constrained CBFs.

Contribution and Novelty: The proposed Probabilistic Collision Cone Control Barrier Function (PCC-CBF) directly addresses this gap. It synthesizes the dynamic efficiency of the deterministic Collision Cone constraint with the rigorous probabilistic framework of Chance-Constrained CBFs. By explicitly modeling relative state uncertainty p_r and v_r as Gaussian random variables within the CC-CBF structure, the PCC-CBF provides a novel, tractable, and robust mechanism for real-time, safety-critical control in uncertain environments.

3 Control Barrier Function

Control Barrier Functions (CBFs) are mathematical constructs used in control theory to ensure the safety of dynamical systems. They provide a systematic way to enforce safety constraints by defining a safe set of states that the system should remain within. CBFs are particularly useful in scenarios where safety is critical, such as in autonomous vehicles, robotics, and aerospace applications. A Control Barrier Function $h(x)$ is a scalar function that maps the state space to the real numbers. The safe set is defined as the set of states where $h(x) \geq 0$. The boundary of the safe set is given by $h(x) = 0$, and states where $h(x) < 0$ are considered unsafe. To ensure that the system remains within the safe set, the CBF must satisfy certain con-

ditions. Specifically, for a dynamical system described by equation (1):

$$\dot{x} = f(x) + g(x)u. \quad (1)$$

Where x is the state, u is the control input, and $f(x)$ and $g(x)$ are system dynamics, the CBF must satisfy the following inequality (2):

$$\dot{h}(x, u) + \alpha(h(x)) \geq 0 \quad (2)$$

Specifying the barrier function $h(x)$ is a critical design choice, as it determines the shape of the safe set. In many robotic applications, simple distance-based formulations (e.g., maintaining $\|p_r\| \geq r$ where r is the minimum separation distance) are used to enforce collision avoidance. While intuitive, such constraints are often overly conservative: they do not account for the relative motion between agents, and may unnecessarily restrict the system's ability to navigate safely in dense or dynamic environments.

To address this limitation, the concept of the *Collision Cone* (CC) has been introduced as a geometric tool to characterize dangerous relative velocities. The CC describes the set of relative velocity vectors that would eventually lead to a collision, given the current relative position between two agents. Incorporating the CC into the barrier function allows one to explicitly capture the interplay between position and velocity, resulting in less conservative and more dynamically appropriate safety constraints.

This motivates the definition of a Collision Cone Control Barrier Function (CC-CBF), which encodes the requirement that the relative motion of the ego vehicle and obstacle remains outside the collision cone. The corresponding formulation is presented below.

Collision cone:

$$h(x) = \langle p_r, v_r \rangle + \|p_r\| \cdot \|v_r\| \cos \phi \quad (3)$$

where $\cos \phi = \frac{\sqrt{\|p_r\|^2 - r^2}}{\|p_r\|}$

[4]

The collision cone is a geometric concept used to predict potential collisions between two moving objects, such as vehicles or robots. It defines the set of relative velocity vectors that, if maintained, will eventually result in a collision given the current positions of the objects.

In more detail: Imagine two agents: an ego vehicle and an obstacle. The relative position between them is p_r , and the relative velocity is v_r . The collision cone is constructed by drawing lines from the ego vehicle to the edges of the obstacle, forming a cone-shaped region in velocity space. If the relative velocity v_r lies inside this cone, the ego vehicle is on a collision course with the obstacle. If v_r is outside the cone, the agents will not collide, assuming their velocities remain constant. By incorporating the collision cone into a control barrier function, the system can enforce safety constraints that are sensitive to both position and velocity, allowing for less conservative and more flexible collision avoidance compared to simple distance-based rules. This approach is especially useful in dynamic environments where both agents may be moving.

Derivative of such defined barrier function has following form:

$$\begin{aligned} \dot{h}(x, u) = & \langle \dot{p}_r, v_r \rangle + \langle p_r, \dot{v}_r \rangle \\ & + \langle v_r, \dot{v}_r \rangle \frac{\sqrt{\|p_r\|^2 - r^2}}{\|v_r\|} \\ & + \langle p_r, \dot{p}_r \rangle \frac{\|p_r\|}{\sqrt{\|p_r\|^2 - r^2}} \end{aligned} \quad (4)$$

This formula will be reformulated in next section to has form as in equation 2.

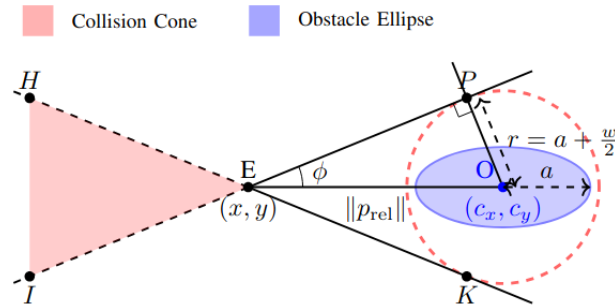


Figure 1: Construction of collision cone for an alliptical obstacle considering the ego vehicle's dimensions (width w).

4 Probabilistic Collision Cone CBF

The Collision Cone Control Barrier Function (CC-CBF) introduced in the previous section

provides a deterministic safety certificate. In particular, the condition (2) ensures that the relative motion between the robot and an obstacle remains outside the collision cone. However, this formulation implicitly assumes perfect knowledge of the obstacle's position and velocity. In real-world scenarios, perception is subject to sensor noise, occlusions, and state estimation errors, which may lead to unsafe behavior if not explicitly accounted for.

To incorporate uncertainty, we model the relative position p_r and velocity v_r of the obstacle as random variables. Let

$$p_r = \hat{p}_r + \delta_p, \quad v_r = \hat{v}_r + \delta_v, \quad (5)$$

where $\hat{p}_r, \hat{v}_r$ denote the estimated values, and δ_p, δ_v are zero-mean Gaussian random variables with covariances Σ_p and Σ_v , respectively:

$$\delta_p \sim \mathcal{N}(0, \Sigma_p), \quad \delta_v \sim \mathcal{N}(0, \Sigma_v). \quad (6)$$

By linearizing the deterministic barrier function around the estimated state $\hat{x}$, we obtain a first-order approximation:

$$h(x) \approx h(\hat{x}) + \nabla_p h(\hat{x})^T \delta_p + \nabla_v h(\hat{x})^T \delta_v \quad (7)$$

where:

- $\delta_p \sim \mathcal{N}(0, \Sigma_p)$ and $\delta_v \sim \mathcal{N}(0, \Sigma_v)$,
- $\nabla_p h(\hat{x}) = v_r + \frac{\|v_r\|}{\Delta} p_r$,
- $\nabla_v h(\hat{x}) = p_r + \frac{\Delta}{\|v_r\|} v_r$,
- $\Delta = \sqrt{\|p_r\|^2 - r^2}$.

Similarly, the derivative $\dot{h}(x, u)$ can be expressed as:

$$\begin{aligned} \dot{h}(x, u) \approx & \dot{h}(\hat{x}, u) \\ & + \nabla_p \dot{h}(\hat{x}, u)^T \delta_p + \nabla_v \dot{h}(\hat{x}, u)^T \delta_v + \nabla_a \dot{h}(\hat{x}, u)^T \delta_a, \end{aligned}$$

where:

- $\nabla_p \dot{h}(\hat{x}, u) = a_r + \frac{\langle v_r, a_r \rangle}{\|v_r\| \Delta} p_r + \frac{\|v_r\|}{\Delta} v_r - \frac{\|v_r\| \langle p_r, v_r \rangle}{\Delta^3} p_r$,
- $\nabla_v \dot{h}(\hat{x}, u) = 2v_r + \Delta \left(\frac{1}{\|v_r\|} a_r - \frac{\langle v_r, a_r \rangle}{\|v_r\|^3} v_r \right) + \frac{1}{\Delta} \left(\|v_r\| p_r + \frac{\langle p_r, v_r \rangle}{\|v_r\|} v_r \right)$,

- $\nabla_a \dot{h}(\hat{x}, u) = p_r + \frac{\Delta}{\|v_r\|} v_r$.

This allows us to represent the safety constraint (2) as a random variable $c_h(x, u)$:

$$\underbrace{\overbrace{\dot{h}(\hat{x}, u) + \alpha(h(\hat{x}, u))}^{c_h(x, u)}}_{\mu_{c_h}} + \delta_{c_h} \geq 0, \quad (8)$$

where:

- μ_{c_h} is the mean of $c_h(x, u)$ random variable,
- δ_{c_h} is a zero-mean Gaussian random variable with variance:
 - $\sigma_{c_h}^2 = \nabla_p^T \Sigma_p \nabla_p^T + \nabla_v^T \Sigma_v \nabla_v^T + \nabla_a^T \Sigma_a \nabla_a^T$,
 - $\nabla_p = \nabla_p \dot{h}(\hat{x}, u) + \nabla_p h(\hat{x}, u)$,
 - $\nabla_v = \nabla_v \dot{h}(\hat{x}, u) + \nabla_v h(\hat{x}, u)$,
 - $\nabla_a = \nabla_a \dot{h}(\hat{x}, u)$.

Instead of requiring $c_h(x, u) \geq 0$ deterministically, we impose a chance constraint:

$$\mathbb{P}[c_h(x, u) \geq 0] \geq 1 - \nu, \quad (9)$$

where $\nu \in (0, 1)$ denotes the acceptable risk level. Under the Gaussian approximation, condition (9) is equivalent to:

$$\mu_{c_h} - \Phi^{-1}(1 - \nu) \sigma_{c_h} \geq 0, \quad (10)$$

where Φ^{-1} is the inverse standard normal cumulative distribution function. This yields a tractable inequality constraint that can be incorporated into the quadratic program used for CBF-based control.

Equation (10) constitutes the proposed *Probabilistic Collision Cone Control Barrier Function (PCC-CBF)*. It ensures that the system maintains safety with probability at least $1 - \nu$, while explicitly accounting for uncertainty in perception.

5 Experiments

Experiments were planned in two steps: first simulations - implemented in MATLAB/Simulink, second stage on real robots.

5.1 Motion Models

Two types of robot models are commonly encountered in industrial practice. The first is the unicycle model, typically used for indoor mobile robots. The second is the bicycle model, which serves as a standard representation for cars in driver assistance systems and autonomous driving applications.

5.1.1 Unicycle Model

The unicycle kinematic model is a simple yet effective representation for differential-drive robots and other mobile platforms that can move forward and rotate, but cannot move sideways. The model captures the essential nonholonomic constraint that the robot's velocity is always aligned with its heading direction.

In this model, the state vector consists of the position (x_p, y_p) of the robot in the plane, its orientation θ , linear velocity v , and angular velocity ω . The control inputs are the linear acceleration a and angular acceleration δ .

The continuous-time kinematic equations for the unicycle model are:

$$\underbrace{\begin{bmatrix} \dot{x}_u \\ \dot{y}_u \\ \dot{\theta}_u \\ \dot{v}_u \\ \dot{\omega}_u \end{bmatrix}}_{\dot{x}_u} = \underbrace{\begin{bmatrix} v_u \cos \theta_u \\ v_u \sin \theta_u \\ \omega_u \\ 0 \\ 0 \end{bmatrix}}_{f_u(x_u)} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_{g_u(x_u)} \underbrace{\begin{bmatrix} a_u \\ \delta_u \end{bmatrix}}_{u_u}. \quad (11)$$

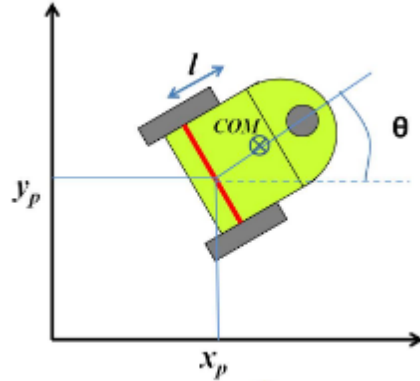


Figure 2: Schematic of unicycle model.

State: The state vector is given by:

$$x_u = [x_u \ y_u \ \theta_u \ v_u \ \omega_u]^T,$$

where (x_u, y_u) represents the position, θ_u the heading angle, v_u the linear velocity, and ω_u the angular velocity.

Control: The control input vector is given by:

$$u_u = [a_u \ d_u]^T,$$

where a_u is the linear acceleration and δ_u is the angular acceleration.

This model is widely used in robotics for motion planning and control due to its simplicity and ability to capture the key motion constraints of many ground vehicles.

5.1.2 Bicycle Model Model

The bicycle model is a widely used kinematic representation for car-like vehicles, capturing the essential nonholonomic constraints and steering dynamics. In this model, the vehicle is abstracted as a single-track system with a front steering wheel and a rear driving wheel. The state vector typically includes the position (x_p, y_p) of the vehicle's reference point (often the rear axle midpoint), the heading angle θ , the longitudinal velocity v , and the yaw rate ω . The control inputs are the longitudinal acceleration a and the steering angle rate δ .

The continuous-time kinematic bicycle model is given by:

$$\underbrace{\begin{bmatrix} \dot{x}_b \\ \dot{y}_b \\ \dot{\theta}_b \\ \dot{v}_b \end{bmatrix}}_{\dot{x}_b} = \underbrace{\begin{bmatrix} v_b \cos(\theta_b) \\ v_b \sin(\theta_b) \\ \frac{v_b}{L} \tan d_b \\ a_b \end{bmatrix}}_{f_b(x_b, u_b)} \quad (12)$$

where L is wheel base, as it is shown in Figure 3,

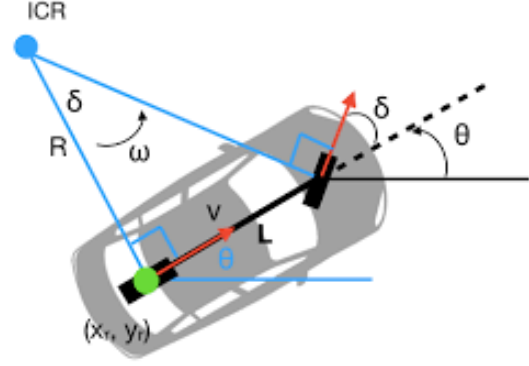


Figure 3: Schematic of bicycle model.

State: The state vector is given by:

$$x_b = [x_b \ y_b \ \theta_b \ v_b]^T,$$

where (x_b, y_b) represents the position, θ_b the heading angle, and v_b the longitudinal velocity.

Control: The control input vector is given by:

$$u_b = [a_b \ d_b]^T,$$

where a_b is the longitudinal acceleration and δ_b is the steering angle rate.

This model captures the coupling between steering and heading, and is commonly used for path planning and control in autonomous driving applications.

5.2 Simulations

5.2.1 Unicycle Model Simulations

The unicycle model was tested in a scenario where two robots are on a collision course. Both robots are controlled by the same controller, which is a combination of a CBF and a nominal controller. Each robot has a goal position to reach, which is located on the opposite side of the other robot. The robots start with access to their own position and velocity, as well as noisy measurements of the other robot's position and velocity. The noise is modeled as zero-mean Gaussian with standard deviation of $0.2m$ and $0.2m/s$ for position and velocity, respectively.

Simulation results, trajectories of both robots, for different values of α (from equation

2) are presented in Figure 4. It can be observed that the robots trajectories are diverge from the straight line produced by the nominal controller, to avoid collision.

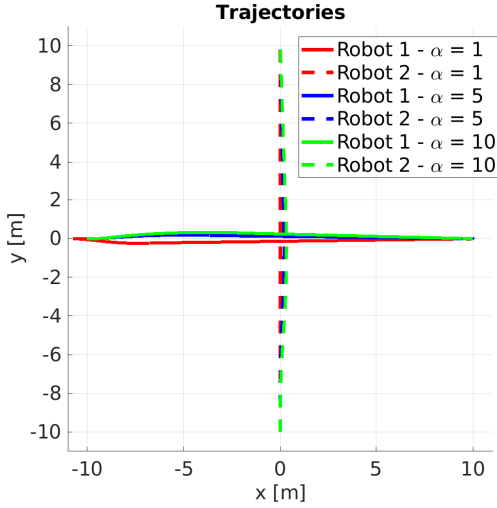


Figure 4: Trajectories of robots with unicycle kinematics for different α values.

The distance between robots during the simulation is presented in Figure 5. As can be seen, the distance between robots is always greater than the safety distance (1 m), which means that the collision is avoided. The distance is larger for smaller values of α , which means that the robots are more conservative in their motion.

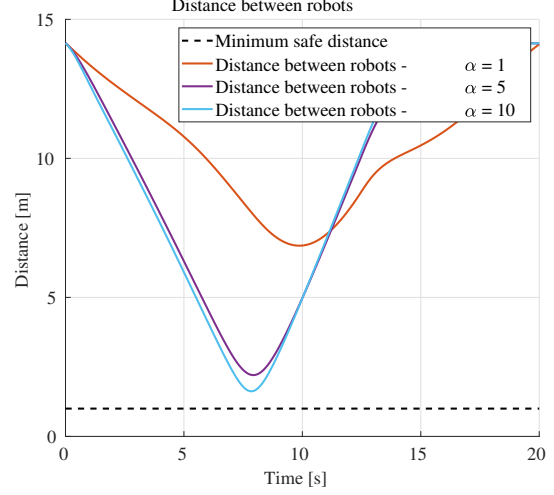


Figure 5: Distance between robots (unicycle model) - it is clearly visible that in any case safe distance is crossed.

Second experiment was conducted with one robot using nominal controller only, and the other using CBF to avoid collision. The position of both robots during the simulation is presented in Figure 6. It can be observed that the robot using CBF is able to avoid collision with the other robot, it starts, stops for a moment, and then continues safely to its goal.

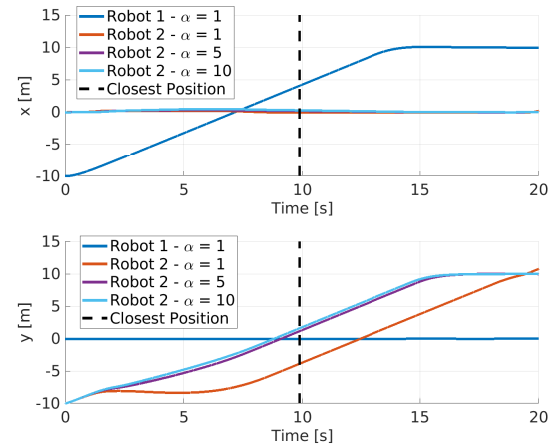


Figure 6: Position of both robots. Robot 1 runs nominal controller only, and Robot 2 uses CBF to avoid collision.

5.2.2 Bicycle Model Simulations

The bicycle model was tested in a similar scenario as the unicycle model. Two robots are on a collision course, each controlled by the same controller, which is a combination of a CBF and a nominal controller. Each robot has a goal position to reach, which is located on the opposite side of the other robot. The robots start have access to own position and velocity, as well as noisy measurements of the other robot's position and velocity. The noise is modeled as zero-mean Gaussian with standard deviation of $0.2m$ and $0.2m/s$ for position and velocity, respectively.

Simulation results, trajectories of both robots, for different values of α (from equation 2) are presented in Figure 7. It can be observed that the robots trajectories are diverge from the straight line produced by the nominal controller, to avoid collision. Those deviations are visibly larger than i case of unicycle model, which is caused by the nonholonomic constraints of the bicycle model. Also this time in case of smallest value of α can be observed waiting by one robot for second robot pass.

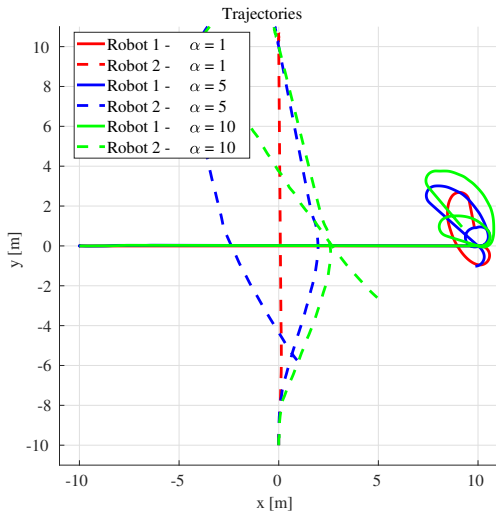


Figure 7: Trajectory of robots with bicycle kinematics for different α values.

In figure 8 distance between robots during the simulation is presented. As can be seen, the distance between robots is always greater than

the safety distance (1 m), which means that the collision is avoided and the PCC-CBF works correctly.

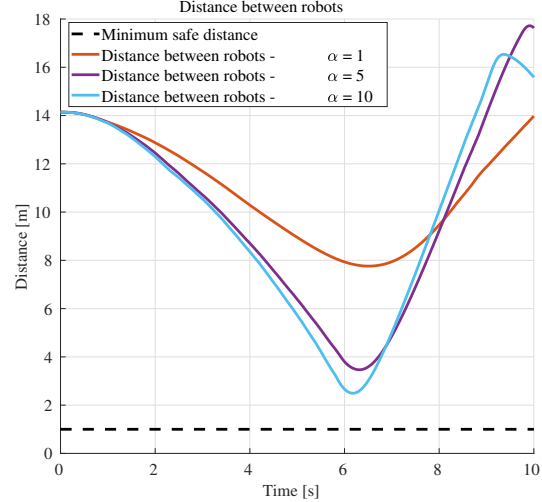


Figure 8: Distance between robots (bicycle model) - it is clearly visible that in any case safe distance is crossed.

5.3 Real Experiments

6 Summary

This paper presented the Probabilistic Collision Cone Control Barrier Function (PCC-CBF) — a novel framework for enforcing safety in autonomous systems operating under perception uncertainty. By combining the geometric concept of the Collision Cone with Control Barrier Functions and extending it to a probabilistic formulation, the proposed method provides formal safety guarantees in the presence of noisy or uncertain state estimates.

The PCC-CBF reformulates the collision avoidance condition as a chance constraint, ensuring that the probability of collision remains below a specified risk threshold. Using Gaussian models of uncertainty in relative position and velocity, we derived a tractable, closed-form safety condition that can be integrated into standard quadratic programming-based controllers for real-time operation.

Simulation studies on unicycle and bicycle models demonstrate that the proposed framework successfully maintains safe trajectories while preserving control performance, even when state estimates are corrupted by noise.

Future work will focus on extending the PCC-CBF framework to multi-agent scenarios, incorporating time-varying uncertainty estimates from Bayesian perception pipelines, and validating the approach in large-scale outdoor experiments.

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